

1. Professor Peach: There are 4 students in a class that lasts 90 minutes. For each student, the average number of questions they ask in a 45 minute time period is 1.

a) Professor Peach hands out a fruit to a student every time they ask a question. However, each student has a limit of 3 fruits. Assume the probability of a student asking a question is independent of another student asking a question and the previous questions they've asked. What is the probability that some student will not get a fruit for every question they ask?

b) There is no longer a limit on the number of fruits per student. However, Professor Peach only has 8 total fruits. What is the probability that there will be exactly enough fruit for each question asked (i.e, exactly 8 questions are asked among the students)? Note that the sum of independent Poisson random variables is Poisson with the sum of the means.

c) What is the probability that Professor Peach will run out of fruit?

d) Suppose that each of the students has preferred fruits; 2 prefer oranges and 2 prefer bananas. Professor Peach has exactly 4 oranges and 4 bananas. If the professor ran out of a student's preferred fruit and the student asks a question, they will get the other type of fruit. What is the probability that given exactly 8 questions were asked, for every question asked, the student got their preferred fruit?

2. Bank Heist: A bank installed a vibration sensor to detect potential thefts. Let X be the value the sensor reads. If there is no theft, $X \sim N(\mu = 0, \sigma = 1)$. If there is a theft, $X \sim N(\mu = 4, \sigma = 1)$. The prior probability of a theft is 0.001. In practice, however, the sensor suffers from an unknown calibration error: the reading is scaled by a factor α . The actual observed reading is $Y = \alpha X$, where $\alpha \sim Uni(0.8, 1.2)$

a). For a single observation y , derive the likelihood function for $P(y|\alpha, theft)$, $P(y|\alpha, notheft)$ and $P(y|\alpha)$.

b) Call the expressions obtained from part a for $P(y|\alpha, theft)$, $P(y|\alpha, notheft)$ and $P(y|\alpha)$ p_1, p_2, p_3 respectively. Using these, calculate $P(y|theft)$, $P(y|notheft)$ and $P(y)$ by considering all the possibilities for α .

c. Express the posterior probability that a theft has occurred given the observation y . Call the values obtained in part b for $P(y|theft)$, $P(y|notheft)$ and $P(y)$ p_{y1}, p_{y2}, p_{y3} respectively.

d. Discuss how would a wider range of alpha with the same expected value would impact the marginal likelihoods $P(y|notheft)$ and $P(y|theft)$. How would it impact the posteriors $P(theft|y)$, $P(notheft|y)$?

3. Typing contest: In a head-to-head typing contest, you and your friend race to complete a series of characters as quickly as possible. You type a character in 200 milliseconds on average. Your friend types a character in 210 milliseconds on average. You have a 5 percent chance of typing the wrong character. Your friend has a 1 percent chance of typing the wrong character. Whenever an error occurs, you must press backspace and retype the correct character. You cannot proceed to the next character until the current one is correctly entered. Once you move on to the next character, your previous character is 'completed'.

a. What is the expected amount of time you spend to complete one character? What about your friend?

b. Denote your answer from part a as e_y, e_f for the expected time for you and your friend respectively. The amount of time it takes you and your friend to hit a character have variances of 10 and 15 milliseconds respectively. What is the probability that you complete one character faster than your friend?

c. What is the probability that you beat your friend in a contest that has 200 characters?

4. College Applications: You are applying to college. You have 10 colleges on your list. 5 have a 10 percent acceptance rate, 3 have a 5 percent acceptance rate, and 2 have a 3 percent acceptance rate..

a. You apply to all colleges on your list. What is the probability that you get into at least one?

b. What is the expected number of colleges you get into?

c. Unfortunately, these colleges are stingy and make you pay a 100 dollar fee to apply. However, you only have a 400 dollar budget. Assuming you pick 4 colleges at random to apply to, what is the probability that you get into at least one?

In parts d and e, you no longer care about your budget

d. In addition to acceptances, your chance of being waitlisted from a specific college is half the chance of being accepted by that same college. For all colleges on your list, if you are waitlisted, you have a 0.1 probability of getting off the waitlist. What is the probability you get into at least one college?

e. If you get waitlisted, the college makes you pay a 10 dollar fee if you still want to be considered. Assume you pay the fee for every college you get waitlisted at, from what is the expected money you spend applying to college?

5. You find yourself in a mystery location on campus with your vision blurred. You know that your current location is one of the following three possibilities: the Engineering Quad, Wilbur Dining, or GovCo. Your prior beliefs about being at each location are: Engineering Quad: $P(EQ) = 0.5$, Wilbur: $P(W) = 0.3$, GovCo: $P(G) = 0.2$. For 5 of your friends, you also have an indicator whether they are nearby. For each friend, the probability they are nearby is independent of the probability that another friend is nearby and depends on your location:

Engineering Quad: Each friend is nearby with probability 0.8.

Wilbur: Each friend is nearby with probability 0.4.

GovCo: Each friend is nearby with probability 0.2. Call this event E .

- Write out the likelihood function $P(E|l)$ for each possible location l .
- Express the posterior probability for each location after observing that exactly 3 of 5 of your friends are nearby.
- Now, suppose that regardless of your location, you are equally likely to observe any count of nearby friends (i.e. 0, 1, 2, 3, 4, or 5) with equal probability. Under this uniform likelihood assumption, mathematically prove that there will not be any update in your belief from your prior belief.

6. **simpleenroll.stanford.edu**: Suppose you want to sign up for the class "PATH51". Unfortunately, this class is in high demand. It only has 300 seats, but 500 students want to take it. Based on factors such as internet, computer speed, etc, the system takes some time to process the request. Assume that the expected amount of time for 10 more students to be enrolled in the class is 10 seconds if there are less than 300 students in the class. Otherwise, enrollment will stop. You will be able to enroll if you are the 300th person to enroll or earlier.

- What is the maximum number of seconds it can take you to enroll so that you have a 90 percent chance of being enrolled?
- You attempt to enroll at time 0, but you have a hold. It takes you 60 seconds to get rid of the hold and try enrolling again, and 20 seconds for the system to process your request. Assume you are the only student with this issue. What is the probability that you are able to enroll?
- Assume you and exactly k other students have a hold, where you and all k other students take exactly 60 seconds to get rid of the hold and try enrolling again. What is the probability that you are able to enroll?

d. Assume the same situation as c, but the amount of other students that have a hold is uniform among 0 and 499. What is the probability you are able to enroll?

Solutions

1a. Each student has a limit of 3 fruits. That is, if a student asks 4 or more questions then the student will not get a fruit from the 4th question onward. For student i , denote the number of questions they ask as $Q_i \sim Poi(2)$. For a given student, the probability the ask 0, 1, 2, or 3 questions is $P(Q_i \leq 3) = \sum_{q=0}^3 \frac{e^{-2}2^q}{q!}$. Because the students are independent, the probability that every student asks at most 3 questions is $(\sum_{q=0}^3 \frac{e^{-2}2^q}{q!})^4$. Thus the probability that some student asks more than 3 questions (and so does not get a fruit for every question) is $1 - (\sum_{q=0}^3 \frac{e^{-2}2^q}{q!})^4$

1b. Now there is no per-student limit; instead the professor has 8 fruits total. All questions across the 4 students are independent, and the total number of questions $N = \sum_{i=1}^4 Q_i$ is poisson with lambr 2+2+2+2=8, as the sum of independent Poisson random variables is Poisson with the sum of the means. Thus, the probability that exactly 8 questions are asked is $P(N = 8) = \frac{e^{-8}8^8}{8!}$

1c. The professor “runs out of fruit” if the total number of questions exceeds 8. (If exactly 8 questions are asked, there is just enough fruit.) Therefore, the probability of running out is $P(N > 8) = 1 - P(N \leq 8) = 1 - \sum_{n=0}^8 \frac{e^{-8}8^n}{n!}$.

1d. We begin by noting that each of the 4 students asks questions according to a Poisson process. Over the 90-minute class, each student’s mean is 2 (since the rate is 1 per 45 minutes). For the 2 orange preferring students, the total number of questions is $X \sim Poi(4)$, and for the 2 banana preferring students, the total number of questions is $Y \sim Poi(4)$. The professor has exactly 4 oranges and 4 bananas. A student gets their preferred fruit provided that the professor’s supply of that fruit is not exceeded. In other words, for the group of orange-preferring students, if they ask more than 4 questions, then after the 4th question a substitution will occur. (Similarly for the banana-preferring group.) Therefore, for every question to be answered with the preferred fruit, we must have $X \leq 4, Y \leq 4$ Since $X+Y = 8$ the only way is for $X = Y = 4$. By the PMF of a poisson random variable, $P(X = 4) = P(Y = 4) = \frac{e^{-2}2^4}{4!}$. Since X and Y are independent, $P(X = 4) \wedge P(Y = 4) = P(X = 4) \cdot P(Y = 4) = (\frac{e^{-2}2^4}{4!})^2$.

2a. A key observation is that $X = Y/\alpha$. If there is no theft, $X \sim N(\mu = 0, \sigma^2 = 1)$, so $Y \sim N(\mu = 0, \sigma^2 = \alpha)$. If there is theft, $X \sim N(\mu = 4, \sigma^2 = 1)$, so $Y \sim N(\mu = 4\alpha, \sigma^2 = \alpha)$

To find $P(Y = y|\alpha, theft)$ and $P(Y = y|\alpha, notheft)$, we use the PDF for a normal distribution.

$$\text{When no theft occurs, } P(y|\alpha, notheft) = \frac{1}{\alpha\sqrt{2\pi}} e^{-\frac{y^2}{2\alpha^2}}$$

When theft occurs, $P(y|\alpha, theft) = \frac{1}{\alpha\sqrt{2\pi}} e^{-\frac{(y-4\alpha)^2}{2\alpha^2}}$ Since theft happens 0.001 if the time, no theft happens 1-0.001 = 0.999 of the time. Since either theft occurs or no theft occurs, $P(y|\alpha) = 0.001P(y|\alpha, theft) + 0.999P(y|\alpha, notheft)$.

2b. Since α can take on any value in the range $[0.8, 1.2]$, we must integrate over the range. Thus,

$$P(y|theft) = \int_{0.8}^{1.2} p_1 d\alpha$$

$$P(y|notheft) = \int_{0.8}^{1.2} p_2 d\alpha$$

$$P(y) = \int_{0.8}^{1.2} p_3 d\alpha$$

2c. Using Bayes' theorem, the probability that a theft is taking place given the observation $P(theft|y) = \frac{P(y)P(y|theft)}{P(theft)}$. In our notation, this becomes:

$$P(theft|y) = \frac{p_{y1} \cdot 0.001}{p_{y3}}. \text{ Similarly, the probability that no theft is occurring is:}$$

$$P(notheft|y) = \frac{p_{y2} \cdot 0.999}{p_{y3}}.$$

2d. Suppose we widen the range of α while keeping the same expected value. On the Marginal Likelihoods $P(y—theft)$ and $P(y—notheft)$, with a broader range of α , there is more uncertainty about the sensor's calibration. In forming the marginal likelihood by integrating over a larger interval, the sharp peaks present when α is nearly known will be “averaged out.” In other words, the resulting marginal likelihoods become more spread out (or “flatter”), and their maximum values are lower because the integration now covers many values of α that are not near the “ideal” value for explaining y . On the Posterior Probabilities $P(theft—y)$ and $P(notheft—y)$, since these posteriors are given by the ratio of the marginal likelihoods (weighted by the prior probabilities), if the effect of widening the α range is similar for both hypotheses then the discrimination provided by the observation y is reduced. That is, the sensor reading y will have less “pull” in moving your beliefs away from the priors. In the limit where the calibration is very uncertain, the posterior probabilities will tend to be closer to the prior probabilities (0.001 for theft and 0.999 for no theft).

3a. To “complete” one character, you must eventually type the correct letter. In doing so, you will make a number N of attempts. In each attempt you type the character (one key press). If the attempt is wrong you also must press backspace (another key press) before trying again. Thus, if you take N attempts the total number of key presses is $M = 1 + 2(N-1) = 2N - 1$. Note that n is a geometric random variable with success probability p , so $E(N) = 1/p$. Thus, $E(M) = 2/p - 1$

Your mean time per key press is 200 ms. Hence your expected time to “clear” one character is:

$$E_{you} = 200(2/0.95 - 1)\text{ms}$$

Similarly, your friend's time per key press is 210 ms and $p=0.99$. Thus:

$$E_{friend} = 210(2/.99 - 1)\text{ms}$$

3b. The overall time per character is a normal random variable:

$$\text{Your time: } T_y \sim N(e_y, 10)$$

$$\text{Your friend's time: } T_f \sim N(e_f, 15).$$

We want to find $P(T_y < T_f)$, which is equivalent to computing $P(T_y - T_f < 0)$. Define $D = T_y - T_f$. Then, $D \sim N(e_y - e_f, 10 + 15 = 25)$. Then, using normal CDF, $P(D < 0) = \phi\left(\frac{0 - (e_y - e_f)}{\sqrt{25}}\right) = \phi\left(\frac{e_f - e_y}{5}\right)$.

3c. In a contest with 200 characters, let total time for you be $S_y = \sum_{i=1}^{200} T_{y_i}$ and let the total time for your friend be $S_f = \sum_{i=1}^{200} T_{y_f}$. Since the times for individual characters are assumed independent, the sums will also be normally distributed:

Your total time: $S_y \sim N(200e_y, 200 * 10)$

Your friend's total time: $S_f \sim N(200e_f, 200 * 15)$

Define the difference in your total times as $D_{200} = S_y - S_f \sim N(200(e_y - e_f), 200(10 + 15) = 5000)$

We want the probability that you beat your friend, i.e. $P(D_{200} < 0)$. Using the Normal CDF, we obtain $\phi\left(\frac{0 - 200(e_y - e_f)}{\sqrt{5000}}\right)$

4a. You get in if at least one college accepts you. For each college the “rejection” probability is one minus its acceptance probability. That is: For Type 10 percent acceptance: $1 - 0.10 = 0.90$, For 0.05 percent acceptance: $1 - 0.05 = 0.95$, and for 0.03 percent acceptance: $1 - 0.03 = 0.97$. Since you apply to all 10 colleges, the probability that none admit you is the product of the individual rejection probabilities: $0.9^5 \cdot 0.95^3 \cdot 0.97^2$. Thus, the probability of at least one acceptance is $1 - 0.9^5 \cdot 0.95^3 \cdot 0.97^2$

4b. Because the admissions decisions are independent, the expected number of acceptances is just the sum of the acceptance probabilities over all colleges. For each type:

10 percent acceptance: $5 \cdot 0.1 = 0.5$

5 percent acceptance: $3 \cdot 0.05 = 0.15$

3 percent acceptance $2 \cdot 0.03 = 0.06$

Thus, the overall expected number is: $0.50 + 0.15 + 0.06 = 0.71$, So on average, you expect about 0.71 acceptances.

4c. Because the 10 colleges are not identical, we need to average over the different possible compositions of the 4-college subset. One way is to “partition” the 4 choices by type. Suppose in your chosen set you have:

x 10 percent acceptance rate colleges

y 5 percent acceptance rate colleges

z 3 percent acceptance rate colleges, where $x + y + z = 4$ and $0 \leq x \leq 5, 0 \leq y \leq 3, 0 \leq z \leq 2$.

For a given triple (x,y,z), the probability that none of those colleges accept you is: $0.9^x 0.95^y 0.97^z$. The number of ways to pick such a triple is $\binom{5}{x} \binom{3}{y} \binom{2}{z}$,

Since there are $\binom{10}{4}$ equally likely subsets, the overall probability of no acceptance is: $\frac{1}{104} \sum_{x+y+z=4, x \leq 5, y \leq 3, z \leq 2} \binom{5}{x} \binom{3}{y} \binom{2}{z} 0.9^x 0.95^y 0.97^z$. Then the probability that you get into at least one college is $= 1 - P(\text{no acceptance})$, which was

evaluated above.

4d. We evaluate the new probabilities of eventually getting into a specific college by considering getting accepted in the first round and getting off the waitlist.

10 percent acceptance: $0.1 + 0.1 \cdot 0.5 \cdot 0.1 = 0.105$

5 percent acceptance: $0.05 + 0.05 \cdot 0.5 \cdot 0.1 = 0.0525$

3 percent acceptance: $0.03 + 0.03 \cdot 0.5 \cdot 0.1 = 0.0315$

Thus, for each college the probability of not getting in is:

10 percent acceptance: $1 - 0.105 = 0.895$

5 percent acceptance: $1 - 0.0525 = 0.9475$

3 percent acceptance: $1 - 0.0315 = 0.9685$

Then, the probability of getting rejected from all is $0.895^5 0.9475^3 0.9685^2$, So the probability that you get into at least one college is: $1 - 0.895^5 0.9475^3 0.9685^2$

4e. For a college with acceptance probability p , the waitlist probability is $p/2$, so the extra expected fee at that college is $10(p/2) = 5p$ dollars. Thus, the expected cost per college is: $100 + 5p$. Now, summing over the 10 colleges by type: 10 percent (5 colleges) = $100 + 5(0.10) = 100 + 0.5 = 100.5$.

Total: $5 \cdot 100.5 = 502.5$

5 percent (3 colleges) = $100 + 5(0.05) = 100 + 0.25 = 100.25$.

Total: 300.75

3 percent (2 colleges) = $100 + 5(0.03) = 100 + 0.15 = 100.15$.

Total: 200.30

Adding these up, the overall expected money spent is: $502.5 + 300.75 + 200.30 = 1003.55$ dollars.

5a. Because the number of nearby friends follows a Binomial distribution with $n=5$ and location-specific success probability p , we have: $P(E|l) = \binom{5}{3} p^3 (1-p)^2$. Thus, $P(E|EQ) = \binom{5}{3} 0.8^3 (0.2)^2$, $P(E|W) = \binom{5}{3} 0.4^3 (0.6)^2$, and $P(E|G) = \binom{5}{3} 0.2^3 (0.8)^2$

5b. By bayes theorem, $P(l|E) = \frac{P(E|l)P(l)}{P(E)} = \frac{P(E|l)P(l)}{\sum_{l'} P(E|l')P(l')}$. Plugging in the priors, we obtain the following posteriors:

$$P(EQ|E) = \frac{\binom{5}{3} 0.8^3 (0.2)^2 \cdot 0.5}{\binom{5}{3} 0.8^3 (0.2)^2 \cdot 0.5 + \binom{5}{3} 0.4^3 (0.6)^2 \cdot 0.3 + \binom{5}{3} 0.2^3 (0.8)^2 \cdot 0.2}$$

$$P(W|E) = \frac{\binom{5}{3} 0.4^3 (0.6)^2 \cdot 0.3}{\binom{5}{3} 0.8^3 (0.2)^2 \cdot 0.5 + \binom{5}{3} 0.4^3 (0.6)^2 \cdot 0.3 + \binom{5}{3} 0.2^3 (0.8)^2 \cdot 0.2}$$

$$P(G|E) = \frac{\binom{5}{3} 0.2^3 (0.8)^2 \cdot 0.2}{\binom{5}{3} 0.8^3 (0.2)^2 \cdot 0.5 + \binom{5}{3} 0.4^3 (0.6)^2 \cdot 0.3 + \binom{5}{3} 0.2^3 (0.8)^2 \cdot 0.2}$$

5c. Now suppose that regardless of your location, every possible count of nearby friends (0, 1, 2, 3, 4, or 5) is equally likely. Define E_i as the event of exactly i friends being nearby. Then $P(E_i|l) = 1/6$ for all l . Thus, the updated posterior becomes $P(l|E_i) = \frac{P(E_i|l)P(l)}{P(E_i)}$. Since by assumption $P(E_i|l) = 1/6$ for every location l , the numerator becomes $\frac{1}{6}P(l)$ and the marginal likeli-

hood is $P(E_i) = \sum_{l'} P(E_i|l')P(l') = \sum_{l'} \frac{1}{6}P(l') = \frac{1}{6} \sum_{l'} P(l') = 1/6$. Thus, $P(l|E_i) = \frac{\frac{1}{6}P(l)}{\frac{1}{6}} = P(l)$. In other words, under the uniform likelihood assumption, the number of friends near you provides no additional information, so the posterior remains exactly the same as the prior.

6a. When you finish at time t , each other student has probability $p(t) = e^{-t/20}$ to have finished before you. (There are 499 “other” students.) Thus the number $X \sim \text{Bin}(499, p(t))$ of other students finishing before time t must be ≤ 299 in order for you to be admitted. We therefore wish to choose $t = t^*$ such that $P(X \leq 299) = \sum_{j=0}^{299} \binom{499}{j} [1 - e^{-t/20}]^j [e^{-t/20}]^{499-j} = 0.9$

6b. Because you are delayed, your enrollment “finishes” at time $60+20=80$ s. Meanwhile every other student (all 499 of them) starts at time 0 with processing times $T \sim \text{Exp}(1/20)$. For any one of these, the probability of finishing by time 80 is $1 - e^{-80/20} = 1 - e^{-4}$. Thus the number of others finishing by time 80 is $Y \sim \text{Bin}(499, 1 - e^{-4})$. We want to find $P(Y \leq 299)$. To do this, we approximate Y using a normal distribution and calculate the z-score for 299.5. This is equivalent to $z = \frac{299.5 - \mu}{\sigma} = \frac{299.5 - 499(1 - e^{-4})}{\sqrt{499(1 - e^{-4})(e^{-4})}}$. To find $P(Y \leq 299)$, we take $\phi(z) = \phi\left(\frac{299.5 - 499(1 - e^{-4})}{\sqrt{499(1 - e^{-4})(e^{-4})}}\right)$

6c. We can break this problem in to 2 groups: The non-delayed group and the delayed group

Non-delayed group: There are $500 - (k + 1) = 499 - k$ students who start at time 0. For each, the probability to finish by time 80 is $1 - e^{-80/20} = 1 - e^{-4}$. Let $X \sim \text{Bin}(499 - k, 1 - e^{-4})$ denote the number of these students that finish by time 80.

Delayed group: There are $k+1$ students (you plus k others) and all finish exactly at time 80. Since they “tie” at time 80, we assume that the order among them is decided by random tie-breaking.

Your admission rule: You have a chance of being admitted if at most 299 non-delayed students finish before you. The number of non-delayed students finishing by time 80 is $Y \sim \text{Bin}(500 - k - 1, 1 - e^{-4})$. We want to find $P(Y = x)$ for $0 \leq x \leq 299$. To do this, we approximate Y using a normal distribution and calculate the PDF for each x . Then, given that exactly x students are able to enroll before you in the non-delayed group, there are $300 - x$ spots left. The probability that you will get one of these spots is $\min(1, \frac{300 - x}{k + 1})$. Thus, summing over all possibilities of x that give you a chance of enrollment,

$$P(\text{admitted}|k) = \sum_{x=0}^{299} \binom{499-k}{x} (1 - e^{-4})^x (e^{-4})^{499-k-x} \cdot \min(1, \frac{300-x}{k+1})$$

6d. Now you must average the probability from part (c) over all possible values of k . That is, $P(\text{admitted}) = \frac{1}{500} \sum_{k=0}^{499} P(\text{admitted}|k)$, where $P(\text{admitted}|k)$ is derived in part c.